

Executive Summary of Project 1

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Assignment Overview

The purpose of this assignment was to develop a good understanding of how TCP and UDP client-server distributed systems function. The assignment aimed to teach the mechanics of both protocols in terms of how they receive client requests, perform necessary operations, create responses, and send those responses back. It helped clarify the key differences between TCP and UDP, such as the reliability and connection-based nature of TCP versus the connectionless, faster, and less reliable nature of UDP. The scope of the assignment involved implementing a key-value store using a HashMap with operations like "get", "put", and "delete", as well as handling potential errors during the connection and operation execution phases. It also involved correctly logging the responses sent back and forth between the client and server with clear timestamps. As a whole, this project was a first step towards creating our own distributed system which can be handled using multiple packet transfer protocols.

Technical Impression

Overall, my experience with this assignment was both productive and educational. Working with UDP presented some unique challenges due to its connectionless nature, which was quite different from the TCP protocol that I am more familiar with. The absence of a three-way handshake in UDP required me to redesign the client-server communication design, and this was a useful learning experience. I also learned the value of logs and useful timestamps to understand the flow and functioning of the system.

Handling errors, especially socket timeouts and malformed requests, was another valuable aspect of the assignment. It required me to consider various failure scenarios and ensured that I implemented error-handling mechanisms that were both comprehensive and robust. This process also helped me improve my understanding of the libraries and syntax used in both TCP and UDP communication. In addition, implementing the key-value store operations allowed me to apply the knowledge practically and see how data can be handled across distributed systems in real-life scenarios too. I also gained a better understanding of the importance of choosing the right protocol based on the application's needs. Lastly, it helped me hone my Java skills as a lot of good java concepts had to be applied to get the functionality through and implemented efficiently.

Use Case (Application)

A use case for the Key-Value store implemented in this project can be configuration management in systems where a variety of settings can be implemented. In the real world, Key-value stores are used to manage configuration settings in software systems. The settings can be stored with a key

representing the setting's name and the value being its configuration data. So similar PUT, GET and DELETE operations can be executed to manipulate this configuration store as well.